# Eugenio Dellepiane

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## Summary

I am a mathematician currently in my second postdoctoral position. My research interests lie in mathematical analysis, especially in complex analysis and operator theory. I am passionate about teaching and I am actively involved in the international mathematical community: I regularly attend conferences, workshops, and seminars, where I value the opportunity to exchange ideas and and build new collaborations. This CV was updated on October 30th, 2026.

## Education & Research

Sep 2026 – ongoing
**Postdoc at University of Twente (NL)**
Under the supervision of Felix Schwenninger.

Nov 2025 – Aug 2026
**Postdoc at Université Laval, in Québec City (CA)**
Under the supervision of Javad Mashreghi. This position was supposed to last until December 2026. It ended sooner because of teaching duties in the new position at Twente.

Mar – Aug 2025
**Postdoctoral scholarship at Politecnico di Torino (IT)**
I won a five months scholarship to work on the research project titled *"Hardy subspaces, their structure and computational analysis"*, under the supervision of Anita Tabacco.

Oct 2021 – Jan 2025
**Ph.D. in Mathematics at Università di Milano - Statale (IT).**
Thesis title: *An analysis of de Branges–Rovnyak and Dirichlet spaces.*

Sep 2019 – Oct 2021
**M.Sc. in Pure Mathematics at Università di Genova (IT).**
Grade: 110/110 cum laude.
Thesis title: *Singular integrals and functional calculus for divergence-type operators*.

Sep 2016 – Jul 2019
**B.Sc. in Mathematics at Università di Genova (IT).**
Grade: 110/110 cum laude.
Thesis title: *Haar measures on topological groups and the continuous wavelet transform*.

## Research publications

### Journal articles

#### 2026

C. Bellavita, E. Dellepiane, and G. Stylogiannis, "Rhaly operators on weighted Hardy spaces and factorable matrices," *Journal of Geometric Analysis (accepted)*,

E. Dellepiane and D. Seco, "Orthogonal polynomials in de Branges-Rovnyak spaces," *J. Math. Anal. Appl.*, vol. 565, no. 1, Paper No. 130974, 20, 2027. DOI: https://doi.org/10.1016/j.jmaa.2026.130974

C. Bellavita, E. Dellepiane, A. Hartmann, and J. Mashreghi, "Infinitely supported harmonically weighted Dirichlet spaces which are de Branges–Rovnyak spaces," *J. Funct. Anal.*, vol. 291, no. 8, Paper No. 111558, 47 pp. 2026. DOI: https://doi.org/10.1016/j.jfa.2026.111558

E. Dellepiane, J. Mashreghi, M. Nasri, and W. Verreault, "The Cesàro operator on local Dirichlet spaces," *Indag. Math. (N.S.)*, vol. 37, no. 5, pp. 1883–1896, 2026. DOI: https://doi.org/10.1016/j.indag.2026.02.005

**2025**

- C. Bellavita, E. Dellepiane, and G. Stylogiannis, "Boundedness, compactness and Schatten class for Rhaly matrices," *J. Lond. Math. Soc.*, vol. 112, no. 4, Paper no. e70304, 36 pp, 2025. DOI: https://doi.org/10.1112/jlms.70304
- E. Dellepiane, M. M. Peloso, and A. Tabacco, "On the equality of de Branges–Rovnyak and Dirichlet spaces," *Complex Anal. Oper. Theory*, vol. 19, no. 119, 2025. DOI: https://doi.org/10.1007/s11785-025-01747-x
- C. Bellavita, E. Dellepiane, and J. Mashreghi, "The spectral analysis of the difference quotient operator on model spaces," *Bull. Sci. Math.*, vol. 204, no. Paper no. 103673, 22 pp. 2025. DOI: https://doi.org/10.1016/j.bulsci.2025.103673
- C. Bellavita and E. Dellepiane, "Embedding model and de Branges–Rovnyak spaces in Dirichlet spaces," *Complex Var. Elliptic Equ.*, vol. 70, no. 2, pp. 228–246, 2025. DOI: https://doi.org/10.1080/17476933.2023.2298837

### Preprints

- F. J. Cruz-Zamorano, E. Dellepiane, and D. Seco, *Compact and Hilbert–Schmidt composition operators on weighted Hardy spaces*, 2026. arXiv: 2610.40148 [math.FA]. URL: https://arxiv.org/abs/2610.40148
- C. Bellavita, E. Dellepiane, and S. Luo, *A characterization of equality between de Branges–Rovnyak and Dirichlet spaces*, 2026. arXiv: 2610.36943 [math.FA]. URL: https://arxiv.org/abs/2610.36943
- E. Dellepiane, A. Monguzzi, and M. Monti, *Hardy spaces of discrete holomorphic functions on the upper half-lattice*, 2026. arXiv: 2607.17726 [math.CV]. URL: https://arxiv.org/abs/2607.17726

## Conferences & Seminars

### Invited speaker

- **VI Young Researchers Workshop in Harmonic Analysis**
  Università di Milano - Bicocca, 11 - 13 Jan 2027
  I am invited to give a 30 minutes talk.
- **International Workshop on Operator Theory and its Applications (IWOTA26)**
  Université Laval, 3 - 7 August 2026
  I was invited to give two 25 minutes talk, during the special sessions "Operator theory and function spaces" and "Invariant subspaces and cyclicity".
- **CRM-Montreal-Québec Analysis Seminar**
  Université Laval, 14 Nov 2025
  I was invited to give a 1h seminar titled *"Boundedness, compactness and Schatten class for Rhaly matrices"*.
- **Seminario de Análisis Matemático y Matemática Aplicada**
  Universidad de La Laguna, 16 Oct 2025
  I was invited to give a 1h seminar titled *"The difference quotient operator on model spaces"*. I gave the talk in spanish.
- **Séminaire d'Analyse**
  Université de Bordeaux, 2 Oct 2025
  I was invited to give a 1h seminar titled *"Infinitely supported harmonically weighted Dirichlet spaces which are de Branges–Rovnyak spaces"*.
- **Workshop in Complex and Harmonic Analysis**
  Università di Milano - Statale, 16 - 17 September 2025
  I was invited to give a 50 minutes talk, titled *"Difference quotient operators and embeddings between spaces of analytic functions"*.

## Conferences & Seminars (continued)

- **International Workshop on Operator Theory and its Applications (IWOTA25)**
  University of Twente, 14 - 18 July 2025
  I was invited to give a 25 minutes talk, during the special session on operator theory on function spaces in memory of N. Vasilevski, titled *"Boundedness, compactness and Schatten class for Rhaly matrices"*.
- **5th Young Researchers in Functional Analysis and Mathematical Physics meeting**
  University of Aveiro, 8 - 11 July 2025
  I was invited to give a 20 minutes talk, titled *"Boundedness, compactness and Schatten class for Rhaly matrices"*.
- **Analysis on Complex and Discrete Structures**
  Università di Milano - Statale, 20 - 21 February 2025
  I was invited to give a 1h talk titled *"Orthonormal bases in model spaces"*.
- **IV Young Researchers Workshop in Harmonic Analysis**
  Università di Milano - Statale, 22 - 24 Jan 2025
  I was invited to give a 30 minutes talk titled *"Hadamard multipliers and polynomial approximation in Dirichlet spaces"*.
- **Séminaire "Analyse fonctionnelle"**
  Université de Lille, 4 Oct 2024
  I was invited to give a 1h talk titled *"Difference quotient operators and embeddings between spaces of analytic functions"*.
- **III Young Researchers Workshop in Harmonic Analysis**
  Università di Genova, 10 – 12 Jan 2024
  I was invited to give a 30 minutes talk titled *"Embedding model and de Branges-Rovnyak spaces in Dirichlet spaces"*.

### Contributed talks

- **Canadian Mathematical Society Winter Meeting 2025**
  Toronto, 5 - 8 Dec 2025
  I gave a contributed talk titled *"Boundedness, compactness and Schatten class for Rhaly matrices"*.
- **Advanced Courses in Operator Theory and Complex Analysis (ACOTCA25)**
  Université Clermont Auvergne, Clermont Ferrand, 16 - 20 Jun 2025
  I gave a contributed talk titled *"$H(b)$ spaces that are $\mathcal{D}_\mu$ spaces, with infinitely many atoms"*.
- **XLIV Convegno Nazionale di Analisi Armonica**
  Università di Napoli Parthenope, 27 - 30 May 2025
  I gave a contributed talk titled *"Boundedness, compactness and Schatten class for Rhaly matrices"*.
- **Advanced Courses in Operator Theory and Complex Analysis (ACOTCA24)**
  Universidad de La Laguna, Tenerife, 24 – 28 Jun 2024
  I gave a contributed talk titled *"On the difference quotient operator on model spaces"*.
- **Spaces of Analytic Functions and their Operators**
  Saarland University, Saarbrücken, 13 – 17 May 2024
  I gave a contributed talk titled *"Embedding model and de Branges-Rovnyak spaces in Dirichlet spaces"*.
- **Symposium in Harmonic and Complex Analysis, Microlocal and Geometrical Analysis**
  Università di Bologna, 24 – 26 Jan 2024
  I gave a contributed talk titled *"Embedding Model Spaces in Dirichlet spaces"*.

### Others

- **Complex Analysis and Spaces of Holomorphic Functions**
  Würzburg, 21 - 25 Sep 2026

## Conferences & Seminars (continued)

- **Complex Analysis and Operator Theory: a Series of 3-Hour Lecture**
  Milano, 7 - 11 Sep 2026
- **Workshop on Operator Theory, Complex Analysis and Applications (WOTCA25)**
  University of Aveiro, 7 - 11 July 2025
- **Hilbert Function Spaces 2025**
  Villa Mondragone (Frascati), 30 Jun - 4 July 2025
- **Operators on analytic function spaces**
  CIRM - Luminy (Marseille), 2 – 6 Dec 2024
- **Advanced Courses in Operator Theory and Complex Analysis (ACOTCA23)**
  Aristotle University, Thessaloniki, 26 – 30 Jun 2023
- **XLII Convegno Nazionale di Analisi Armonica**
  Università di Milano - Bicocca, 29 – 31 May 2023
- **Young Researchers Workshop in Harmonic Analysis**
  Politecnico di Torino, 19 – 20 Jan 2023
- **(New trends in) complex and Fourier analysis, and operator theory**
  Workshop INdAM in Rome, Italy, 12 – 16 Sep 2022
- **11th International Conference on Harmonic Analysis and Partial Differential Equations**
  ICMAT conference at El Escorial, Madrid, 6 – 10 Jun 2022
- **Geometric Aspects of Complex and Harmonic Analysis**
  Università di Bologna, 17 – 18 Jan 2022
- **8th European Congress of Mathematics**
  Portorož, Slovenia, 19 – 25 Jun 2021

## Teaching

Feb 2027 – May 2027 — **Lecturer**
University of Twente
I am responsible for lectures and tutorials for the course "Mathematics 3" in the degree of Creative Technology.

Nov 2026 – Feb 2027 — **Lecturer**
University of Twente
I am responsible for lectures and tutorials for the course "Mathematics 2" in the degree of Creative Technology.

Sep 2026 – Feb 2027 — **Teaching assistant**
University of Twente
I am running the tutorials for the course "Applied Functional Analysis" in the Master degree in Mathematics.

Sep 2024 – Feb 2025 — **University Tutor**
Università di Milano - Statale
Assisted with exercises for the course "Functional Analysis" in the Master degree in Mathematics.

## Teaching (continued)

Sep 2023 – Feb 2024
**University Tutor**
Università di Milano - Statale
Assisted with exercises for the course "Analisi Matematica 3" in the undergraduate degree in Mathematics.

**University Tutor**
Università di Milano - Statale
Assisted with exercises for the course "Analisi Matematica 1" in the undergraduate degree in Physics.

Feb 2024 – Jun 2024
**University Tutor**
Università di Milano - Statale
Assisted with exercises for the course "Analisi Matematica 2" in the undergraduate degree in Physics.

Sep 2022 – Feb 2023
**Teaching assistant**
Politecnico di Milano
Prepared and taught exercise classes for "Analisi Matematica 1" in the undergradute degree in Mechanical Engineering.

Feb 2022 – Jun 2022
**University Tutor**
Politecnico di Milano
Assisted with exercises for the course "Analisi Matematica 2" in the undergraduate degree in Electrical Engineering.

Mar 2021 – Jul 2021
**University Tutor**
Università di Genova
Assisted students with exercises and mock exams for the course "Matematica generale" in the undergraduate degree in Natural Sciences.

## Miscellaneous

### Periods abroad

Apr – Jun 2024 — **Visiting period (Ph.D.)**. I spent three months in Université Laval in Québec City (CA), working with Javad Mashreghi.

Sep 2020 – Feb 2021 — **Erasmus (Master)**. I spent a semester in Universidad Complutense de Madrid, where I passed 4/4 exams in Spanish language.

Jan – Jun 2019 — **Erasmus (Bachelor)**. I spent a semester in Universidad Complutense de Madrid, where I passed 4/4 exams in Spanish language.

### Languages

- Italian C2 (native language)
- Spanish C1
- English C1
- French B2

## Miscellaneous (continued)

### Ph.D. courses and summer schools

Mar 2025 — **Bounded and unbounded operators on Hilbert spaces**
25 hours Ph.D. course held at Politecnico di Torino by Alessio Martini on spectral theory for bounded and unbounded operators on Hilbert spaces, including the theory of commutative $C^*$-algebras.

Nov 2024 — **Modern singular integral theory: sparse domination**
12 hours Ph.D. course held at Politecnico di Torino by José Conde Alonso (Universidad Autónoma, Madrid).

May 2023 — **Function spaces on the disk and the plane: Model and de Branges spaces, recent results, extensions, and applications**
Two weeks of advanced courses in Cortona, Tuscany, at the Palazzone of the Scuola Normale Superiore. Classes held by William T. Ross and Marco M. Peloso.